

AYUSH RANJAN

San Francisco, CA, USA | aranjan1@ucsc.edu | (831) 266-5973
[linkedin.com/in/ayuranjan](https://www.linkedin.com/in/ayuranjan) | github.com/ayuranjan | ayuranjan.github.io

SUMMARY

Backend Engineer with **3.5+** years of experience, including **2.5 years in production backend engineering** and **1 year in applied AI and agent development**. Experienced in building and operating **distributed microservices at Airbnb** and **monolithic enterprise backend systems at Mercedes-Benz**. In academic labs and independent projects, developed **LLM-based agents, vector search systems, multimodal AI pipelines, and MCP-compatible developer tools** for code and repository understanding. Strong background in **backend architecture, distributed systems, databases, and AI-augmented infrastructure**.

EDUCATION

University of California, Santa Cruz

Sep 2023 – Aug 2025

Master of Science (MS) in Computer Science CGPA: 3.92/4

- **Relevant Coursework:** Analysis of Algorithms, Design and Implementation of Database Systems, Deep Learning for Advanced Computer Vision, Artificial Intelligence (AI), Applied ML: Deep Learning (DL), Computer Networks
- **Teaching Assistant Roles:** **4x TA for Database Systems (CSE-180/182):** led labs/projects on **SQL, transaction management, indexing, stored functions, PL/pgSQL, ETL workflows, and query optimization**. **1x TA for Software Engineering (CSE-115A):** mentored student teams on **Agile development**, version control, and team collaboration practices.

Manipal University, Jaipur

July 2017 – May 2021

Bachelor of Technology (B.Tech.) in Information Technology

- **Relevant Coursework:** Operating Systems, Data Mining and Warehousing, Data Science, Cryptography and Network Security, Advanced Data Structures, Natural Language Processing, Advance Machine Learning Techniques

WORK EXPERIENCE

Altimetrik (Client: Airbnb)

San Francisco, CA, USA

Backend Developer | Dropwizard, Kafka, Thrift RPC, MySQL, Redis, Bazel, K8s

Sep 2025 – Present

- Engineered high-concurrency **Java microservices** for Airbnb Payments Incentives, processing **2.55B requests/month (avg 5.06k QPS)** using **Kafka, Thrift RPC, Dropwizard, and Kubernetes**.
- Optimized developer workflows by contributing to internal **Claude Code** capabilities; engineered **custom commands and skills** to automate Payments-specific codebase navigation, analysis, and boilerplate generation.
- Co-architected **Thrift RPC endpoints** that surfaced **\$120M** in dormant user credits and integrated results into automated re-engagement email pipelines.
- Developed a robust event-driven **Producer-Consumer architecture** using **Kafka** and **Tempo** to automate the lifecycle of Airbnb Virtual Credit Cards (VCC), ensuring 100% automated refund reconciliation upon card expiration.

AI Explainability and Accountability (AIEA) Lab

Santa Cruz, CA, USA

Graduate Researcher | Python, Flask, LangGraph, Whisper, Pinecone, Hugging Face

Oct 2024 – August 2025

- Conducted applied research to improve the reliability and explainability of LLM-based university chatbots, focusing on campus-wide use cases such as enrollment, deadlines, housing, and course queries, leading to enhanced user satisfaction.
- Designed and evaluated **10+ advanced RAG workflow** architectures (Classic RAG, Chain of Thought, Agentic RAG, Adaptive RAG, Corrective RAG, RAT RAG) using comprehensive [Ragas](#) evaluation framework.
- **Fine-tuning open-source LLMs** to align with university-specific tone, structure, and factual accuracy, enabling domain-adaptive generation for student and administrative queries.
- Achieved consistent performance improvements across all architectures, **with an average 35-50% enhancement** over baseline RAG systems, where different approaches excelled in specific metrics (**faithfulness, answer relevancy, context precision**) depending on query complexity and domain requirements.
- Developing production deployment pipeline using Docker containerization, Kubernetes orchestration, and FastAPI endpoints with automated CI/CD workflows, load balancing, and monitoring systems for scalable campus-wide chatbot implementation.

Information Retrieval and Knowledge Management Lab, UCSC

Santa Cruz, CA, USA

AI Engineer Intern | Python, Flask, LangGraph, Whisper, Pinecone, Hugging Face

July 2024 – Sep 2024

- Partnered with a **stealth hardware startup** to develop a **0-to-1 multimodal AI agent** for smart wearable devices (camera-integrated earphones), implementing wake word detection, intent classification, and real-time audio-visual processing for calorie estimation, emergency response and video summarization.
- Designed intelligent query routing system with **95% accuracy** in classifying continuous vs. new queries, integrating **Dialogflow for 8+ pre-built workflows** (calorie estimation, contact calling, emergency location services) and custom **LangGraph agents** for open-domain conversations.
- Engineered real-time multimodal data fusion system combining **audio transcription** (Whisper), **computer vision** (food segmentation, depth estimation), and vector similarity search for intelligent fallback routing to **external tools (web search, OCR)** when confidence scores dropped below 0.8 threshold.

- Developed a multi-threaded memory manager to asynchronously encode and cache historical observations (images, transcripts) into vector embeddings using **Hugging Face transformers**, with storage in **Pinecone**.

Capgemini (Client: Mercedes-Benz)

Mumbai, India

Backend Developer | Java, JDBC, SQL, IBM Db2, Data Modelling

July 2021 – Aug 2023

- Maintained **XDIS**, a **SOAP**-based diagnostic tool structured around a three-tier monolithic Java architecture used by **Mercedes-Benz** service teams for vehicle automation and diagnostics.
- Reduced XML migration time by **67% across 50+ service teams** by migrating to a **delta-based Db2 migration** strategy.
- Optimized export testing by creating a wrapper around the Autosar framework and implementing an **XML file import strategy**, **reducing overall testing time by 40%** and speeding up export time for **individual modules by 17% on average**.
- **Headed the Data Modeling Team**, focusing on backend schema evolution for **vehicle network topology change requests** (e.g., ECU reconfigurations, bus architecture edits).
- Achieved **3rd Place at Innocircle 2022**, Mercedes-Benz Internal Innovation Forum by implementing a micro frontend architecture that enabled users to modify their vehicle network topology and review changes, **reducing process time by over 50%**.
- Developed an **AI-assisted validation system** for over 2,500 historical Change Requests by embedding symbolic vehicle network topologies using custom **Word2Vec** and **Sentence-BERT models**, which flagged rare configurations and recommended optimal topologies, improving validation accuracy.

Capgemini

Mumbai, India

Full-Stack Developer Intern | Spring Boot, React, Redux, MySQL, JUnit

Jan 2021 – May 2021

- Developed a full-stack Medical Portal using **Spring Boot** and **React/Redux**, engineering 17 RESTful endpoints documented with **Swagger** to streamline feature delivery and team collaboration.
- Architected a secure data layer using **MySQL** and **JPA/Hibernate**, implementing **Spring Security** (JWT) with role-based access control and achieving high code reliability through automated **JUnit** and **Jest** testing
- Streamlined deployment by containerizing the application with **Docker**, establishing a **GitHub Actions CI/CD** pipeline, and orchestrating local releases via **Kubernetes** (Minikube) to ensure consistent environments for stakeholders and QA.

SELECTED PROJECTS

Fathom : Code-Aware Search Engine | *Python, FastAPI, ChromaDB, Java, Protobuf* | [GitHub](#) Dec 2025 – Jan 2026

- Designed and implemented Fathom, a code-aware search engine combining **semantic (Tree-sitter + sentence-transformer)**, **structural (SCIP)**, and **literal** search behind a unified FastAPI endpoint for LLM agents.
- Built a project-aware indexing manager (**SQLite Librarian**) and a reproducible indexing pipeline producing embeddings in **ChromaDB** and **SCIP** .scip indices, **enabling accurate go-to-definition and semantic retrieval**.
- Architected Fathom as an **MCP-compatible tool** for autonomous coding agents, improving repository grounding and reducing hallucination risk in code synthesis tasks.

PgVector+ | *C/C++, PL/pgSQL, Database Systems, Vector Similarity* | [Ppt](#)

Jan 2024 – Mar 2024

- Designed and built a **custom PostgreSQL extension** on top of pgvector to support hybrid similarity–dissimilarity search and low-level query composition, bridging gaps in vector DB functionality seen in systems like Pinecone and Qdrant.
- Prototyped a **compound_similarity()** operator in **PL/pgSQL** to support similarity search queries like “similar to X, but unlike Y” using cosine and inner-product thresholds.
- Prototyped PL/pgSQL-based **search_similar_vectors()** function to simulate centroid-based multi-query composition and validate set-based similarity retrieval.
- Earned an **A grade** in CSE 215: Design and Implementation of Database Systems for system-level innovation in vector search.

Unveiling Glitches in CLIP | *Hugging Face, Python, Vector Database, Prompt Engineering* | [arXiv](#) Jan 2024 – Mar 2024

- Conducted in-depth analysis of the CLIP model’s image comprehension capabilities. Identified and documented **14 systemic faults**, including **four novel faults**, impacting CLIP’s interpretation of images using **two novel methodologies**.
- Implemented the Discrepancy Analysis Framework (**DAF**) to analyze discrepancies in image similarity rankings between CLIP and **DINOv2** and utilized **OpenAI’s GPT API** to identify and analyze faults systematically. Utilized the Transformative Caption Analysis for CLIP (**TCAC**) approach to evaluate CLIP’s response to transformations applied to images.
- Achieved **A+ grade** in CSE 290D Neural Computation at UCSC for this project.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, C/C++, Go, Rust, SQL

AI/ML Tooling: NLTK, spaCy, transformers library, Sentence-BERT, Word2Vec, MCP Protocols, MLflow, Prompt Engineering

Backend Development: Spring Boot, REST APIs, GraphQL, FastAPI, Flask, PL/pgSQL, JDBC, Node.js, Express.js

Frontend Development: React, Next.js, Redux, Axios, HTML/CSS, Swagger / OpenAPI, JWT, RBAC

Databases: PostgreSQL, MySQL, Oracle, IBM Db2, MongoDB, Redis, Vector Databases (pgvector, Pinecone, Milvus, Qdrant)

Data Engineering: ETL (Java/XML), Liquebase, Pandas, NumPy, Apache Kafka, Apache Spark

DevOps & MLOps: Docker, Docker Compose, Kubernetes, Minikube, Git, GitHub Actions, Jenkins, Maven, Gradle, Terraform

Testing and Automation: JUnit, Mockito, Jest, React Testing Library, Playwright, TDD, BDD

Cloud and Distributed Systems: AWS (EC2, S3, RDS, Lambda, DynamoDB), Amazon Bedrock, Google Cloud(GCP), Azure

Practices & Miscellaneous: Agile/SAFe, CI/CD, SDLC Lifecycle, Prometheus, Grafana, Apache Airflow